

Reviewer Pack — Minimal Local Cohomology in p-Adic Analysis and Resolution P...

Entry 2c787b079303 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Cohomology in p-Adic Analysis and Resolution Proof Length

Entry ID: 2c787b079303 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 01:56:13 UTC

1. Conjecture

Field A (mathematical branch): p-Adic Analysis **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For every satisfiable Boolean formula φ with n variables, the minimal local cohomology rank of the associated p-adic space of the formula's coefficients is linearly correlated with its resolution proof length, such that $h_0(\text{p-AdicSpace}(\varphi)) = \Theta(L(\varphi))$, where $L(\varphi)$ is the resolution proof length of φ .

Rationale (proposer's reasoning):

p-Adic analysis provides a framework for studying arithmetic properties in a non-Archimedean setting, which may reveal new structural insights into Boolean satisfiability problems. The local cohomology rank measures the complexity of p-adic spaces and could potentially serve as a measure of proof length in resolution proofs.

Taxonomy category: p-adic_analysis (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0d537ab13ab9bf64

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for a statistically significant number of n -variable SAT formulas φ generated with random seeds, the correlation coefficient r from linear regression on $(h_0(\text{p-AdicSpace}(\varphi)), L(\varphi))$ exceeds 0.7, and the constant factor b in the linear model $h_0(\text{p-AdicSpace}(\varphi)) = b * L(\varphi)$ is within $\pm 10\%$ of the expected value.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - p-adic analysis AND Boolean satisfiability Resolution Proof Complexity
- minimal local cohomology p-adic space AND resolution proof length - $h_0(p\text{-AdicSpace}(\phi))$
 $0(L(\phi))$ AND satisfiable Boolean formula

Top relevant hits considered: - [http://arxiv.org/abs/math-ph/0512018v2] On Phase Transitions for P -Adic Potts Model with Competing Interactions on a Cayley Tree - [http://arxiv.org/abs/2408.00810v3] p-adic Equiangular Lines and p-adic van Lint-Seidel Relative Bound - [http://arxiv.org/abs/2201.08874v2] Note on p -adic Local Functional Equation - [http://arxiv.org/abs/2512.11324v2] Measurement of the top-quark production cross-section and charge asymmetry at LHCb - [http://arxiv.org/abs/2411.07730v2] Study of the light scalar $a_0(980)$ through the decay $D^0 \rightarrow a_0(980)^- e^+ \nu_e$ with $a_0(980)^- \rightarrow \eta \pi^-$ - [http://arxiv.org/abs/2303.12927v2] Study of the $f_0(980)$ and $f_0(500)$ Scalar Mesons through the Decay $D_s^+ \rightarrow \pi^+ \pi^- e^+ \nu_e$

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_formula(n):
        return ''.join(random.choice('01') for _ in range(2**n - 1))

    def resolution_length(formula):
        # Simplified heuristic to estimate resolution length
        return len(formula) ** 2

    def p_adic_space_rank(formula):
        # Simplified heuristic to estimate p-adic space rank
        return len(formula)

    n_values = [5, 10, 15, 20, 30, 40]
    instances_tested = 0
    total_h0 = 0
    total_L = 0

    for n in n_values:
```

```

    for _ in range(5):
        formula = generate_formula(n)
        h0 = p_adic_space_rank(formula)
        L = resolution_length(formula)
        instances_tested += 1
        total_h0 += h0
        total_L += L

mean_h0 = total_h0 / instances_tested
mean_L = total_L / instances_tested
correlation_coefficient = (instances_tested * sum(h0 * L for h0, L in zip(total_h0, total_L))
                           mean_h0 * total_L) / math.sqrt((instances_tested * sum(h0**2 for h0 in total_h0)
                                                           (instances_tested * sum(L**2 for L in total_L))

conjecture_holds = correlation_coefficient > 0.7
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "correlation_coefficient",
    "metric_value": correlation_coefficient,
    "instances_tested": instances_tested,
    "n_max": max(n_values),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not complete within the expected time frame. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13803
2	preregistration	ollama_remote	glm4:latest	0	0	9934
3	novelty	ollama_remote	glm4:latest	0	0	10714
4	novelty	ollama_remote	glm4:latest	0	0	12692
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18127
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13028
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	31133
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10314
9	judge	ollama_remote	glm4:latest	0	0	60368

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 180113 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/2c787b079303.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/2c787b079303.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/2c787b079303.tar.gz (if generated)